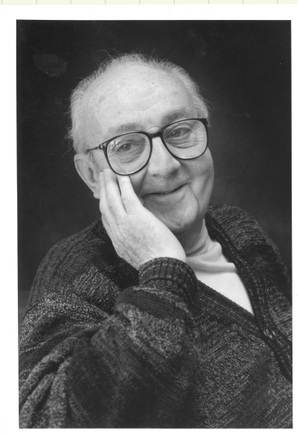
Finite Elemente für Stabtragwerke

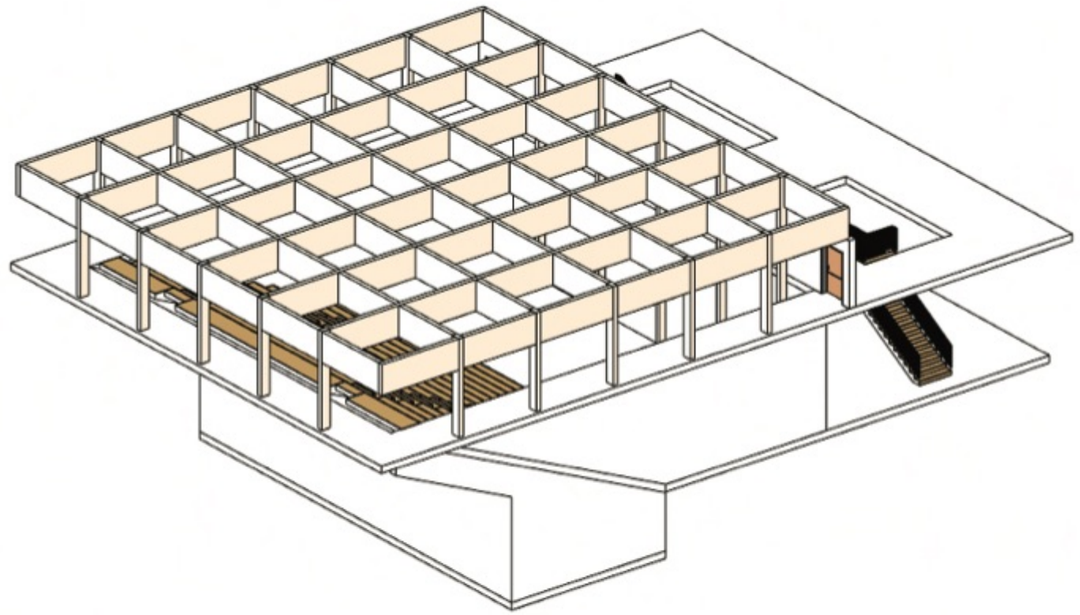
Modellierung von Stabtragwerken

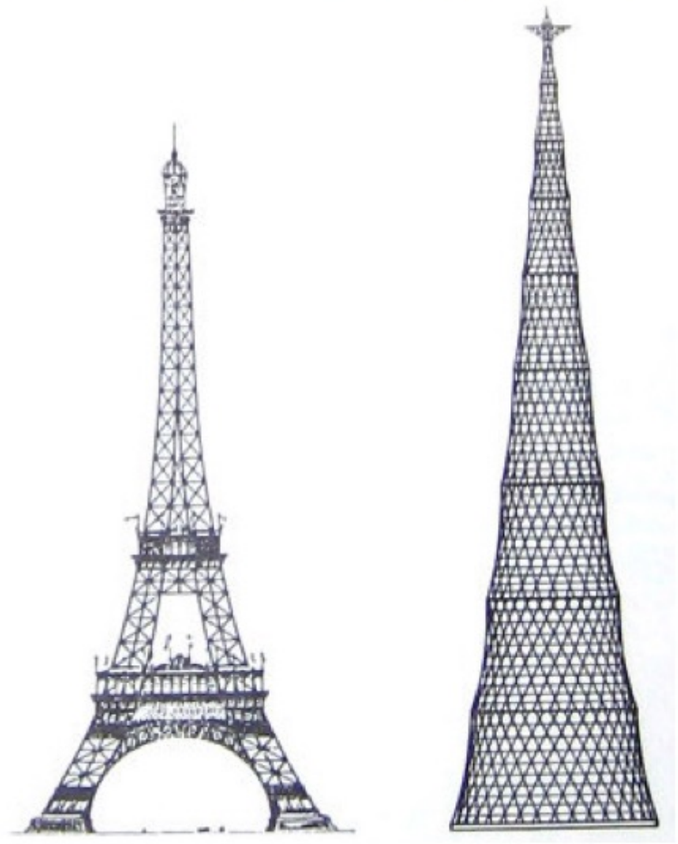
"Essentially, all models are wrong, but some are useful"

George Box, Statistiker



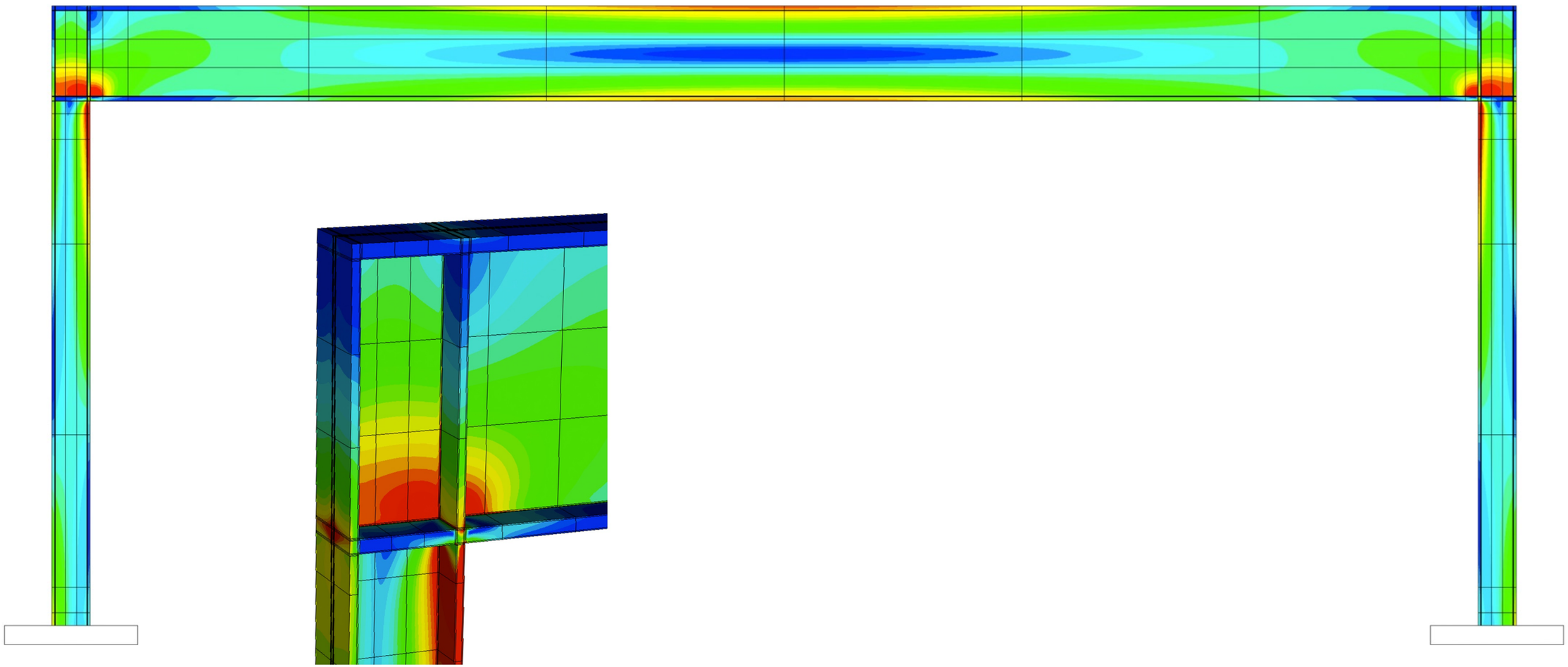




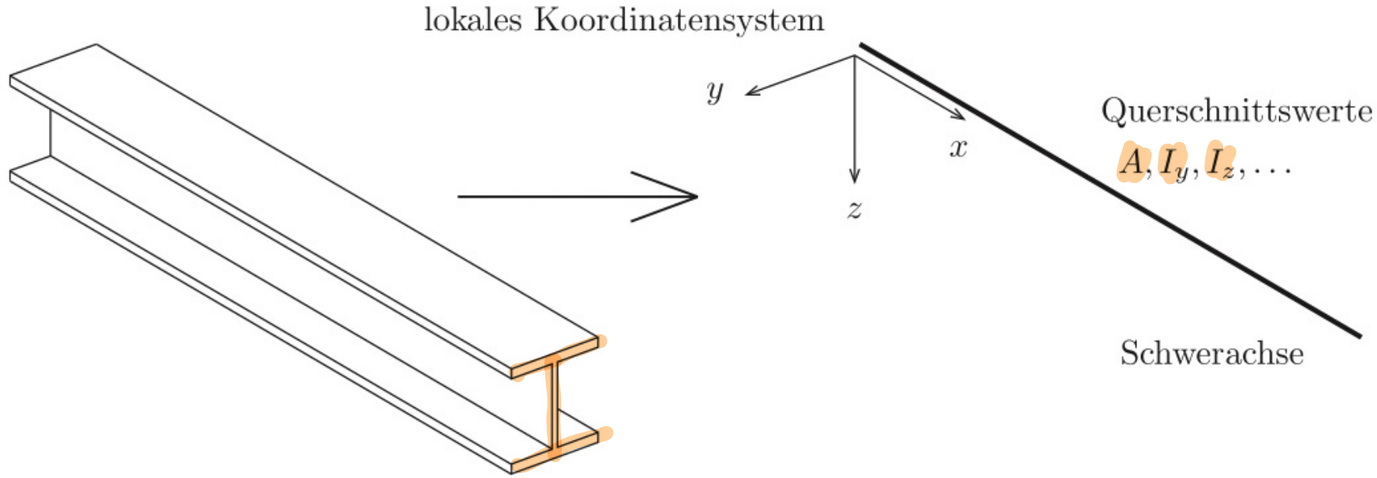


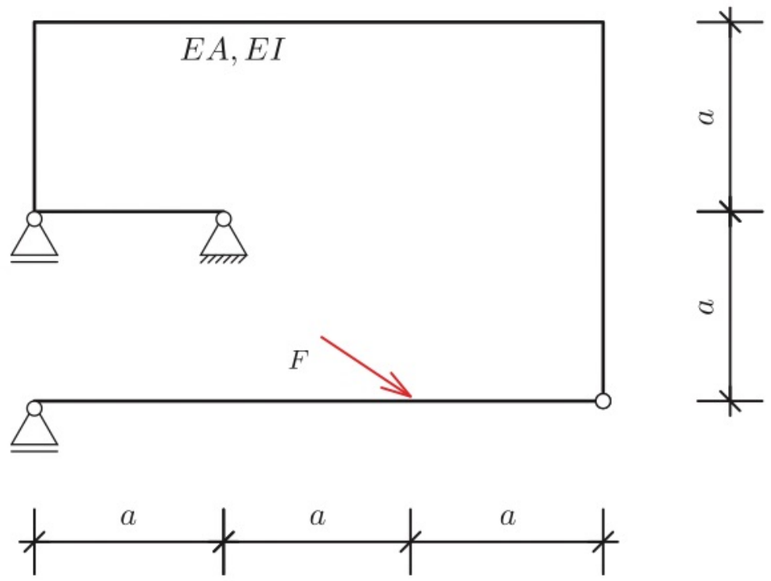


Volumenmodell eines Rahmens

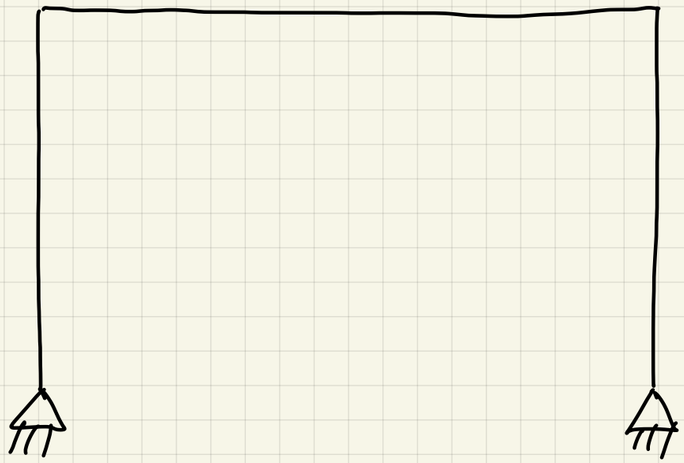


Dimensionsreduzierte Modelle

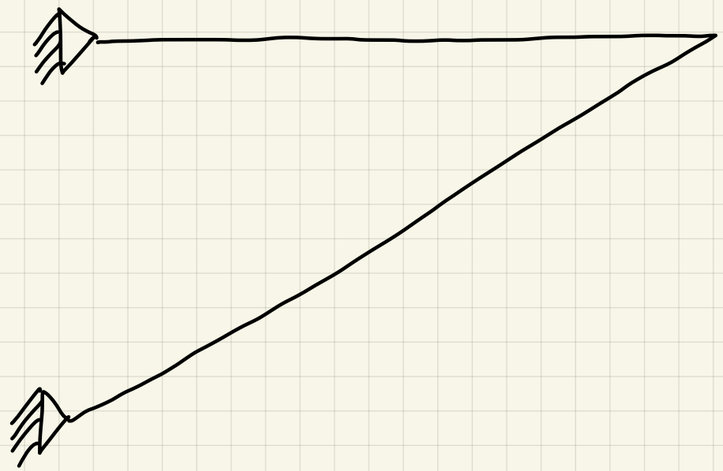




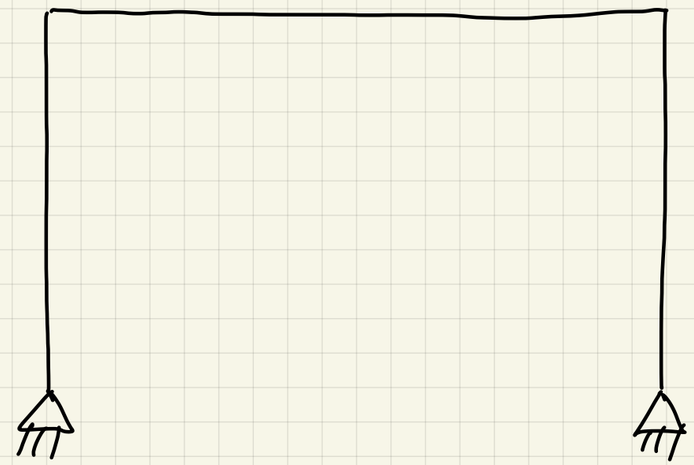
Modell 1: Balkentragwerk

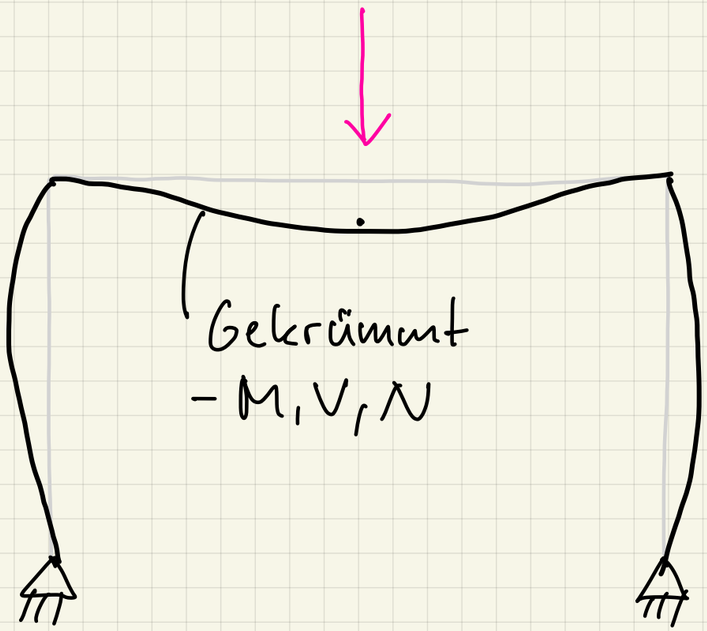


Modell 2: Fachwerk

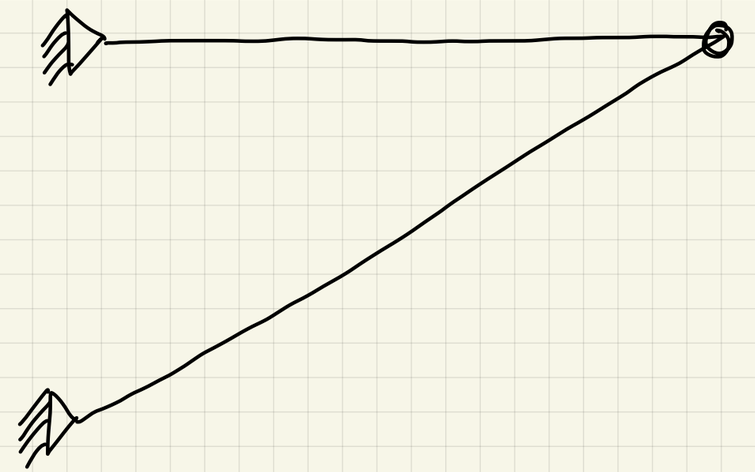


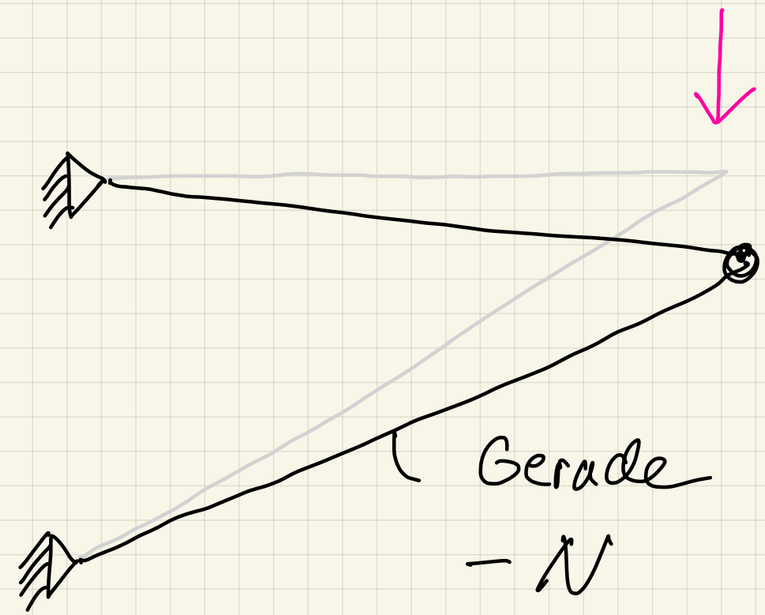
Modell 1: Balkentragwerk





Modell 2: Fachwerk



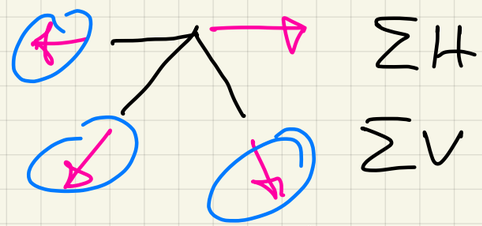




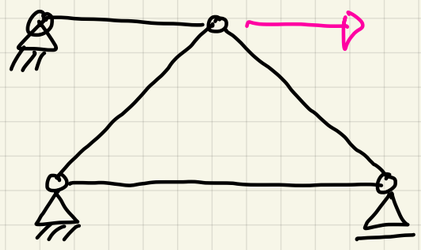




Rechenmethoden für Stabtragwerke

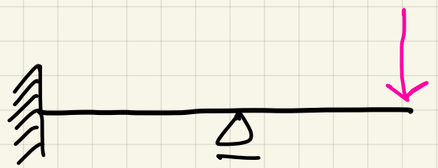


Kräfte zuerst (Schnittprinzip und Gleichgewicht)



+ Handrechnung
+ Anschaulich
− Kompliziert für statisch unbest. Systeme
− Schwer zu formalisieren

Verschiebungen zuerst (Kräfte aus Verformungen)



+ Statische Bestimmtheit spielt keine Rolle
+ Einfach zu formalisieren + programmieren
− Unpraktisch in der Handrechnung
− Wenig anschaulich